

T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 - S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1 S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

Quantity: g_1

$16\pi^2\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$

Column Representative term structure	
T1	$7g_1^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$7g_1^3$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-3g_2^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_2^3$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$

$\beta_{Y_u}^{(1)}$

Quantity: Y_u

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$
T2	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$
T3	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$	$+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure									
T1	$-\frac{3}{2}((Y_d \ (Y_d)^\dagger \ Y_u)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_u)_{ij})$									
T2	$-\frac{9}{4}g_2^2 \ (Y_u)_{ij}$									
T3	$-8g_3^2 \ (Y_u)_{ij}$									
Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d \ (Y_d)^\dagger \ Y_u)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_u)_{ij})$	$-\frac{9}{4}g_2^2 \ (Y_u)_{ij}$	$-8g_3^2 \ (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

Quantity: Y_d

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$
T2	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$
T3	$-8g_3^2\ (Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_d)_{ij}$	$-8g_3^2\ (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Quantity: Y_e

Terms 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$3 \text{Tr} \left(Y_d (Y_d)^\dagger \right) (Y_e)_{ij}$	$+3 \text{Tr} \left(Y_u (Y_u)^\dagger \right) (Y_e)_{ij}$	$+ \text{Tr} \left(Y_e (Y_e)^\dagger \right) (Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure									
T1	$+\frac{1}{2}((h\ (h)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$									
T2	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$									
T3	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$									
Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((h\ (h)^\dagger\ Y_e)_{ij}+3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$	$-\frac{9}{4}g_2^2\ (Y_e)_{ij}$	$-\frac{15}{4}g_1^2\ (Y_e)_{ij}$

$\beta_h^{(1)}$

Quantity: h

$$16\pi^2\,\mu\,\frac{dh}{d\mu}=\beta_h^{(1)}$$

Terms 1 of 2

Column	Representative term structure
T1	$T_\nu(h)_{ij}$
T2	$+\frac{1}{2}((Y_e\ (Y_e)^\dagger\ h)_{ij}+3\left(h\ (h)^\dagger\ h\right)_{ij})$
T3	$-\frac{9}{4}g_2^2\ (h)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$T_\nu(h)_{ij}$	$+\frac{1}{2}((Y_e\ (Y_e)^\dagger\ h)_{ij}+3\left(h\ (h)^\dagger\ h\right)_{ij})$	$-\frac{9}{4}g_2^2\ (h)_{ij}$

Quantity: h

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{3}{4}g_1^2(h)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{4}g_1^2(h)_{ij}$

$\beta_{M_F}^{(1)}$

Quantity: M_F

$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Column	Representative term structure
T1	$\left((h)^T(h)^*M_F\right)_{ij}$
T2	$+ \left((M_F)^T(h)^\dagger h\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\left((h)^T(h)^*M_F\right)_{ij}$	$+ \left((M_F)^T(h)^\dagger h\right)_{ij}$

$\beta_{m_S^2}^{(1)}$

Quantity: m_S^2

$16\pi^2\mu\frac{dm_S^2}{d\mu}=\beta_{m_S^2}^{(1)}$

Terms 1 of 2

Column	Representative term structure
T1	$m_S^2(2T_\nu+6\lambda_2)$
T2	$-8\,\mathrm{Tr}\Big(h\,(M_F)^*\,M_F\,(h)^\dagger\Big)$
T3	$-8\,\mathrm{Tr}\Big(h\,(M_F)^\dagger\,(M_F)^T\,(h)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_S^2(2T_\nu+6\lambda_2)$	$-8\,\mathrm{Tr}\Big(h\,(M_F)^*\,M_F\,(h)^\dagger\Big)$	$-8\,\mathrm{Tr}\Big(h\,(M_F)^\dagger\,(M_F)^T\,(h)^\dagger\Big)$

Quantity: m_S^2

Terms 2 of 2

Column	Representative term structure
T1	$-\frac{9g_2^2m_S^2}{2}$
T2	$-\frac{3g_1^2m_S^2}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_S^2}{2}$	$-\frac{3g_1^2m_S^2}{2}$

Quantity: λ_1

Terms 1 of 5

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_1(12\text{Tr}(Y_d(Y_d)^\dagger) + 4\text{Tr}(Y_e(Y_e)^\dagger) + 12\text{Tr}(Y_u(Y_u)^\dagger))$	$-12\text{Tr}(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger)$	$-4\text{Tr}(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure									
T1	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$									
T2	$+\frac{9g_2^4}{4}$									
T3	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$									
Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$	$+\frac{9g_2^4}{4}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$-9g_2^2\lambda_1$
T2	$+\frac{3g_1^4}{4}$
T3	$+4\lambda_3^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$	$+4\lambda_3^2$

Quantity: λ_1

Terms 4 of 5

Column	Representative term structure
T1	$+4\lambda_3\lambda_4$
T2	$+2\lambda_4^2$
T3	$+2\lambda_5^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_3\lambda_4$	$+2\lambda_4^2$	$+2\lambda_5^2$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+12\lambda_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_1^2$

$\beta_{\lambda_2}^{(1)}$
Quantity: λ_2

$16\pi^2\,\mu\,\frac{d\lambda_2}{d\mu}=\beta_{\lambda_2}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$4\lambda_2 T_\nu$
T2	$-4\,\text{Tr}\Big(h\,(h)^\dagger\,h\,(h)^\dagger\Big)$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_2 T_\nu$	$-4\,\text{Tr}\Big(h\,(h)^\dagger\,h\,(h)^\dagger\Big)$	$+\frac{9g_2^4}{4}$

Quantity: λ_2

Terms 2 of 4

Column	Representative term structure									
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_2\right)$									
T2	$-9g_2^2\lambda_2$									
T3	$+\frac{3g_1^4}{4}$									
Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_2\right)$	$-9g_2^2\lambda_2$	$+\frac{3g_1^4}{4}$

Quantity: λ_2

Terms 3 of 4

Column	Representative term structure
T1	$+4\lambda_3^2$
T2	$+4\lambda_3\lambda_4$
T3	$+2\lambda_4^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_3^2$	$+4\lambda_3\lambda_4$	$+2\lambda_4^2$

Quantity: λ_2

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_5^2$
T2	$+12\lambda_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_5^2$	$+12\lambda_2^2$

Quantity: λ_3

Terms 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_3(2T_\nu + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_4)$	$+\lambda_4(T_\nu + 3\text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3\text{Tr}(Y_u(Y_u)^\dagger))$	$-2\text{Tr}(h(h)^\dagger Y_e(Y_e)^\dagger)$

Quantity: λ_3

Terms 2 of 4

Column Representative term structure											
T1	$+\frac{9g_2^4}{4}$										
T2	$+g_2^2\left(-9\lambda_3-\frac{9\lambda_4}{2}\right)$										
T3	$+\frac{3g_1^4}{4}$										
Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3	
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$+g_2^2\left(-9\lambda_3-\frac{9\lambda_4}{2}\right)$	$+\frac{3g_1^4}{4}$	

Quantity: λ_3

Terms 3 of 4

Column	Representative term structure
T1	$+g_1^2\left(-3\lambda_3-\frac{3\lambda_4}{2}\right)$
T2	$+4\lambda_3^2$
T3	$+\lambda_1(6\lambda_3+3\lambda_4)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_1^2\left(-3\lambda_3-\frac{3\lambda_4}{2}\right)$	$+4\lambda_3^2$	$+\lambda_1(6\lambda_3+3\lambda_4)$

Quantity: λ_3

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_2(6\lambda_3+3\lambda_4)$
T2	$+4\lambda_4^2$
T3	$+6\lambda_5^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_2(6\lambda_3+3\lambda_4)$	$+4\lambda_4^2$	$+6\lambda_5^2$

Quantity: λ_4

Terms 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_3(T_\nu + 3\text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3\text{Tr}(Y_u(Y_u)^\dagger) + 8\lambda_4)$	$+\lambda_4(2T_\nu + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$	$+2\text{Tr}(h(h)^\dagger Y_e(Y_e)^\dagger)$

Quantity: λ_4

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{9g_2^4}{8}$
T2	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_3}{2}-3\lambda_4\right)$
T3	$+g_2^2\left(-\frac{9\lambda_3}{2}-9\lambda_4\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{8}$	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_3}{2}-3\lambda_4\right)$	$+g_2^2\left(-\frac{9\lambda_3}{2}-9\lambda_4\right)$

Quantity: λ_4

Terms 3 of 4

Column	Representative term structure
T1	$+ \frac{3g_1^4}{8}$
T2	$+2\lambda_3^2$
T3	$+\lambda_1(3\lambda_3 + 3\lambda_4)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3g_1^4}{8}$	$+2\lambda_3^2$	$+\lambda_1(3\lambda_3 + 3\lambda_4)$

Quantity: λ_4

Terms 4 of 4

Column	Representative term structure
T1	$+\lambda_2(3\lambda_3+3\lambda_4)$
T2	$+5\lambda_4^2$
T3	$+9\lambda_5^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_2(3\lambda_3+3\lambda_4)$	$+5\lambda_4^2$	$+9\lambda_5^2$

$\beta_{\lambda_5}^{(1)}$
Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5(2T_\nu+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$
T2	$-9g_2^2\lambda_5$
T3	$-3g_1^2\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_5(2T_\nu+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$	$-9g_2^2\lambda_5$	$-3g_1^2\lambda_5$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+8\lambda_3\lambda_5$
T2	$+12\lambda_4\lambda_5$
T3	$+2\lambda_5\lambda_1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_3\lambda_5$	$+12\lambda_4\lambda_5$	$+2\lambda_5\lambda_1$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda_5\lambda_2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_5\lambda_2$